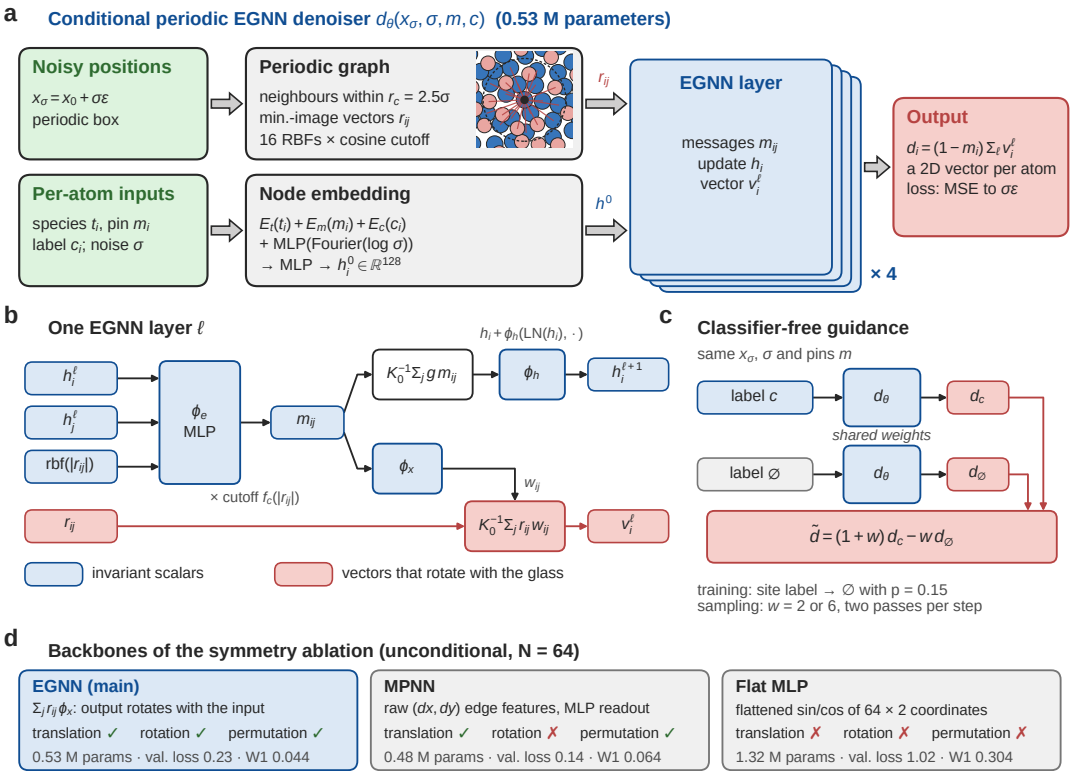

a
Conditional periodic EGNN denoiser $d_\theta(x_\sigma, \sigma, m, c)$ (0.53 M parameters)
Noisy positions
$x_\sigma = x_0 + \sigma\varepsilon$
periodic box
Per-atom inputs
species t_i, pin m_i
label c_i; noise σ
Periodic graph
neighbours within r_c = 2.5σ
min.-image vectors r_{ij}
16 RBFs × cosine cutoff
r_{ij}
Node embedding
$E_t(t_i) + E_m(m_i) + E_c(c_i)$
+ MLP(Fourier(log σ))
→ MLP → $h_i^0 \in \mathbb{R}^{128}$
h^0
EGNN layer
messages m_{ij}
update h_i
vector v_i^ℓ
× 4
Output
$d_i = (1 - m_i)\,\Sigma_\ell\, v_i^\ell$
a 2D vector per atom
loss: MSE to $\sigma\varepsilon$
b
One EGNN layer ℓ
h_i^ℓ
h_j^ℓ
rbf($|r_{ij}|$)
ϕ_e MLP
m_{ij}
× cutoff $f_c(|r_{ij}|)$
$K_0^{-1}\Sigma_j\, g\, m_{ij}$
ϕ_h
$h_i + \phi_h(\mathrm{LN}(h_i), \cdot)$
$h_i^{\ell+1}$
ϕ_x
w_{ij}
r_{ij}
$K_0^{-1}\Sigma_j\, r_{ij}\, w_{ij}$
v_i^ℓ
invariant scalars
vectors that rotate with the glass
c
Classifier-free guidance
same x_σ, σ and pins m
label c
d_θ
d_c
shared weights
label ∅
d_θ
$d_\varnothing$
$\tilde{d} = (1 + w)\, d_c - w\, d_\varnothing$
training: site label → ∅ with p = 0.15
sampling: w = 2 or 6, two passes per step
d
Backbones of the symmetry ablation (unconditional, N = 64)
EGNN (main)
$\Sigma_j r_{ij}\, \phi_x$: output rotates with the input
translation ✓ rotation ✓ permutation ✓
0.53 M params · val. loss 0.23 · W1 0.044
MPNN
raw (dx, dy) edge features, MLP readout
translation ✓ rotation ✗ permutation ✓
0.48 M params · val. loss 0.14 · W1 0.064
Flat MLP
flattened sin/cos of 64 × 2 coordinates
translation ✗ rotation ✗ permutation ✗
1.32 M params · val. loss 1.02 · W1 0.304